



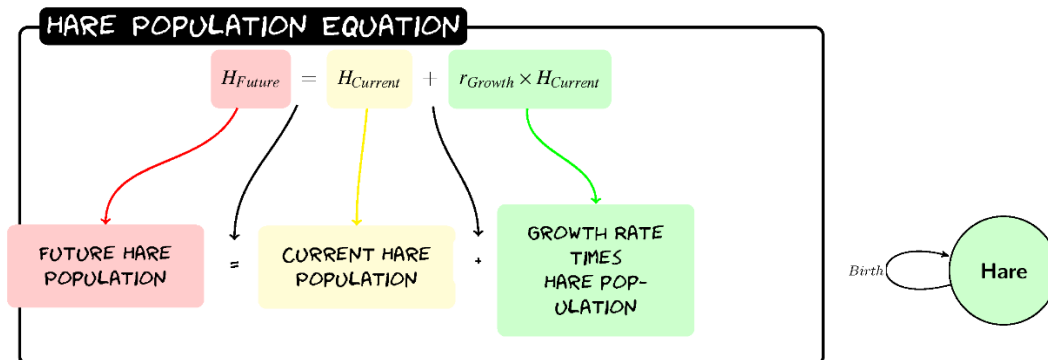
# The Maths of Predators and Preys

## Don't get Eaten by the Wolf

### Hares



No predators



#### Exercise 1: Calculate the Population

1.1 Given a population of 60 Hares and a birth rate of 0.2 what would the population be after one year using the formula:

$$H_{Future} = H + 0.2(H)$$

**Answer:**

$$72 = 60 + 0.2(60)$$

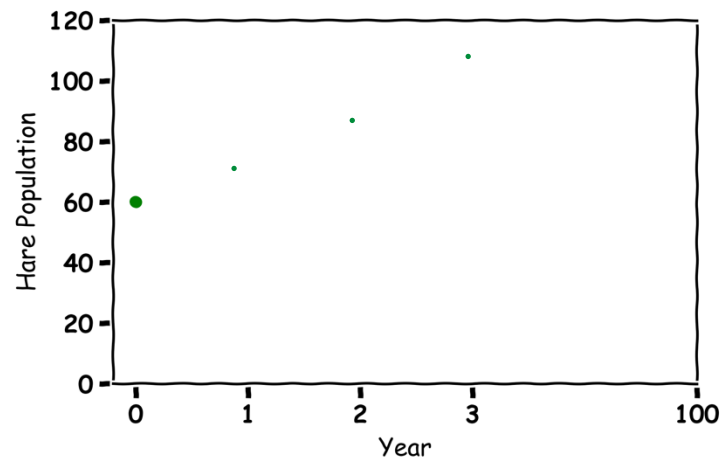
1.2 Now calculate the Hare population for three years

**Answer:**

| Year | Hare Population              |
|------|------------------------------|
| 0    | 60                           |
| 1    | $60 + 0.2 \times (60) = 72$  |
| 2    | $72 + 0.2 \times (72) = 86$  |
| 3    | $86 + 0.2 \times (86) = 103$ |

#### Exercise 2:

Sketch the graph for three years of Hare populations.



### Exercise 3: What would happen in 100 Years

Guess what would happen to the Hare population in 100 years?

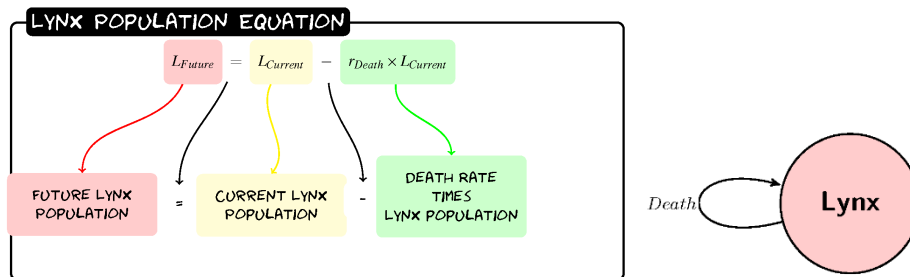
**Answer:**

**Over 4 Billion**



# Lynx

No Prey



4.1 Given a population of 30 Lynxes and a death rate of 0.1 what would the population be after one year?

$$L_{Future} = L - 0.1(L)$$

**Answer:**

$$30 - 0.1(30) = 27$$

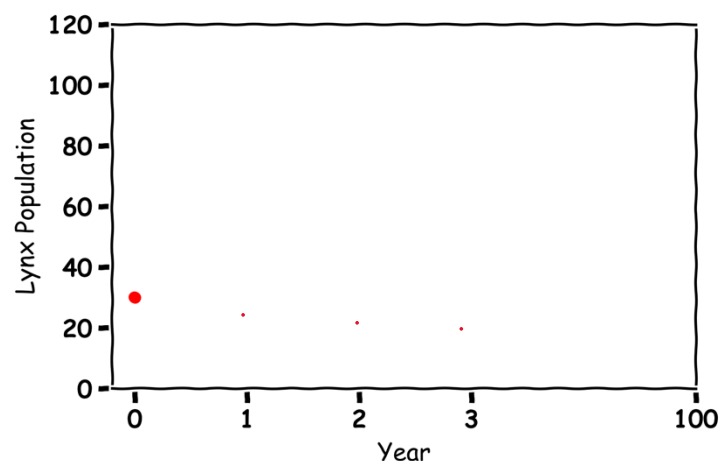
4.2 Now calculate the Lynx population for three years

**Answer:**

| Year | Lynx Population             |
|------|-----------------------------|
| 0    | 30                          |
| 1    | $30 - 0.1 \times (30) = 27$ |
| 2    | $27 - 0.1 \times (27) = 24$ |
| 3    | $24 - 0.1 \times (24) = 22$ |

## Exercise 5

Sketch the graph for three years of Hare populations.

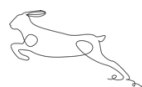


## Exercise 6

Guess what would happen to the Lynx population in 100 years?

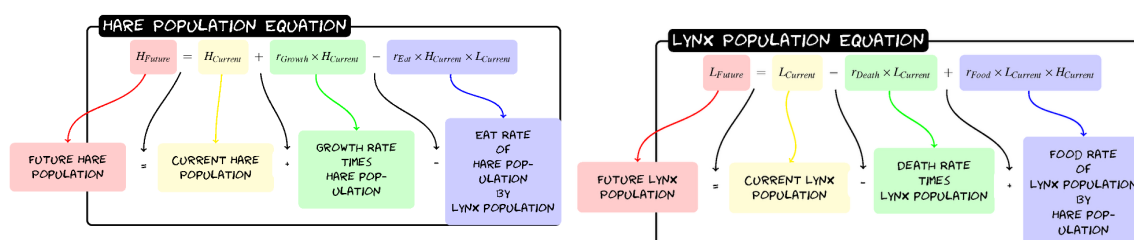
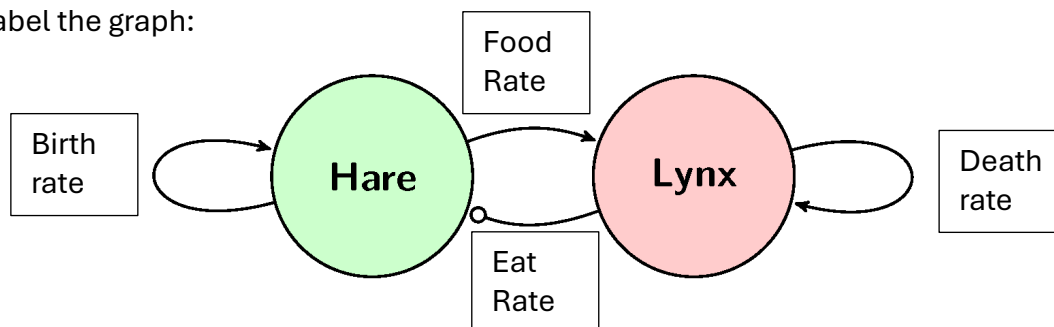
**ANSWER:**

All Dead



## Hares and Lynxes

Label the graph:



### Exercise 8

8.1 Given a population of 60 Hares and 30 Lynxes with a birth rate of 0.2, a death rate of 0.1, a eat rate of 0.005 and a food rate of 0.002 what would the populations be after one year?

| Hare  | Value | Lynx  | Value |
|-------|-------|-------|-------|
| Birth | 0.2   | Death | 0.1   |
| Eat   | 0.005 | Food  | 0.002 |

$$H_{\text{Future}} = H + 0.2(H) - 0.005(H)(L)$$

$$L_{\text{Future}} = L - 0.1(L) + 0.002(L)(H)$$

ANSWER:

HARE:  $60 + 0.2 \cdot (60) - 0.005 \cdot (60) \cdot (30) = 63$

LYNX:  $30 - 0.1 \cdot (30) + 0.002 \cdot (30) \cdot (60) = 31$

8.2 Now Predict Three Years of Hare and Lynx Populations

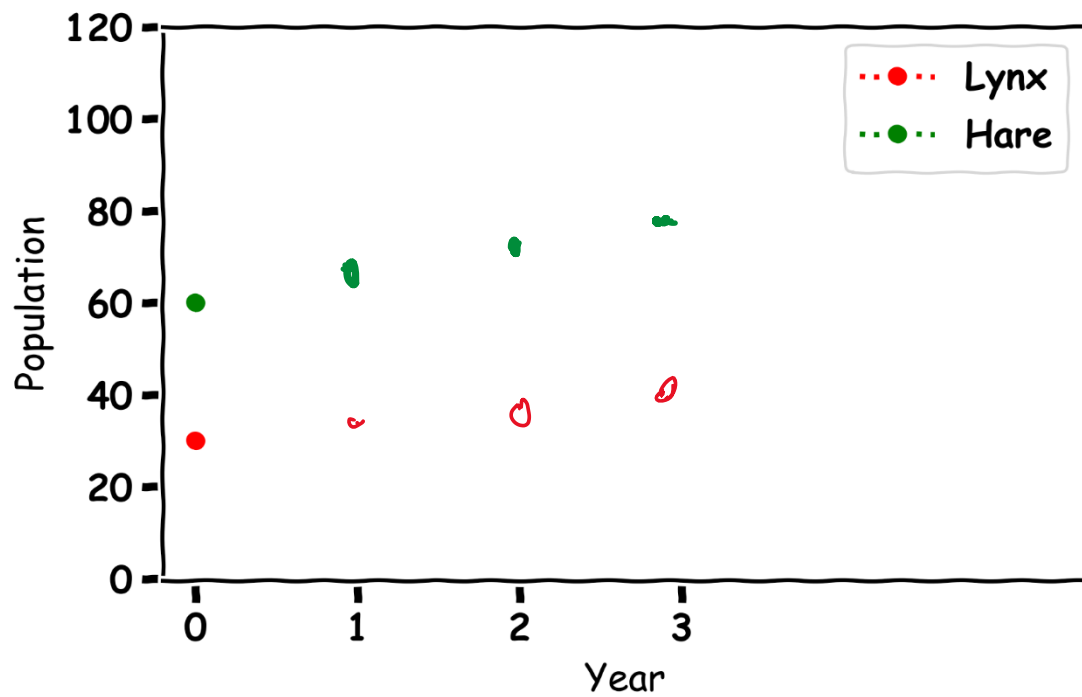
ANSWER:

| Year | Hare Population                                          | Lynx Population                                          |
|------|----------------------------------------------------------|----------------------------------------------------------|
| 0    | 60                                                       | 30                                                       |
| 1    | $60 + 0.2 \cdot (60) - 0.005 \cdot (60) \cdot (30) = 63$ | $30 - 0.1 \cdot (30) + 0.002 \cdot (30) \cdot (60) = 31$ |
| 2    | $63 + 0.2 \cdot (63) - 0.005 \cdot (63) \cdot (31) = 66$ | $31 - 0.1 \cdot (31) + 0.002 \cdot (31) \cdot (63) = 31$ |
| 3    | $66 + 0.2 \cdot (66) - 0.005 \cdot (66) \cdot (31) = 69$ | $31 - 0.1 \cdot (31) + 0.002 \cdot (31) \cdot (66) = 32$ |



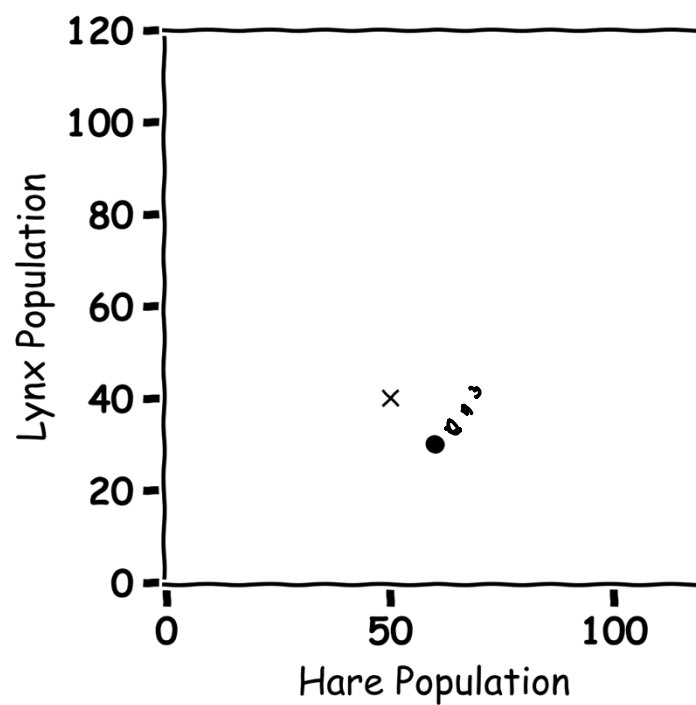
### Exercise 9:

Sketch the graph for the Lynx and Hare populations for three years:



### Exercise 10

Sketch the Hare population on the x-axis and the Lynx population on the y-axis

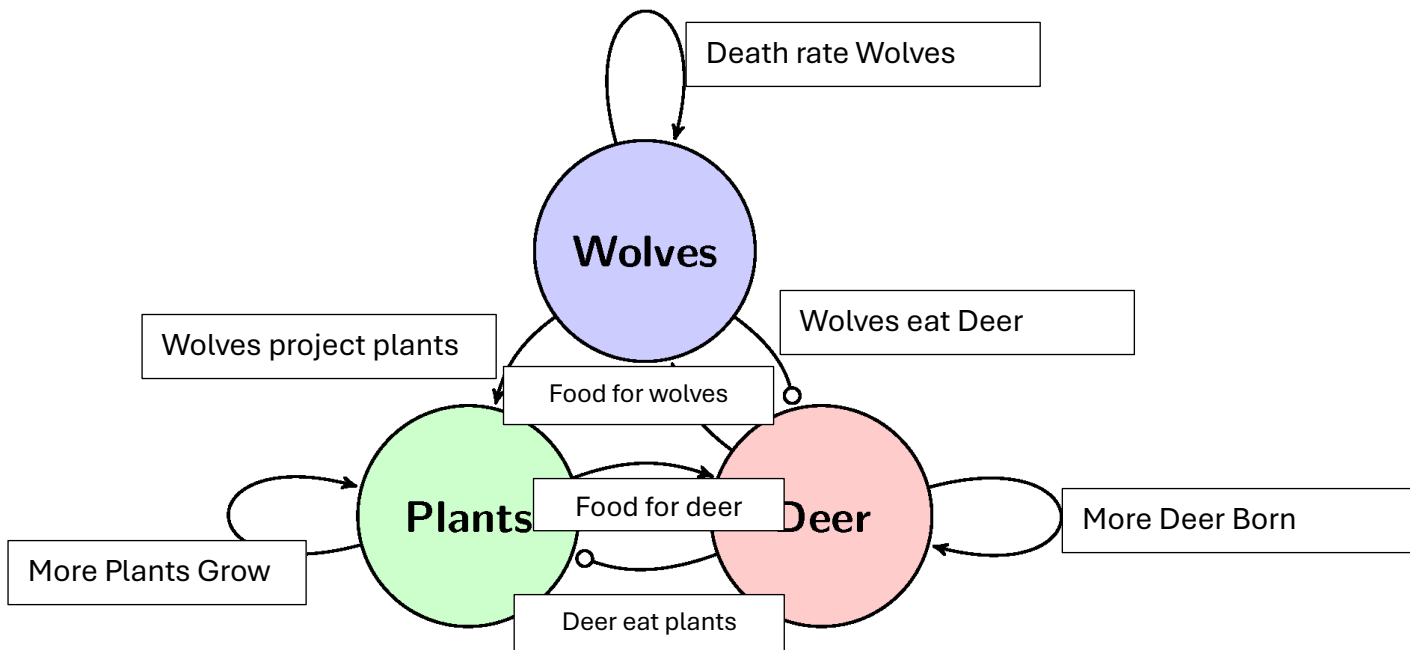




# Don't get Eaten by the Wolf

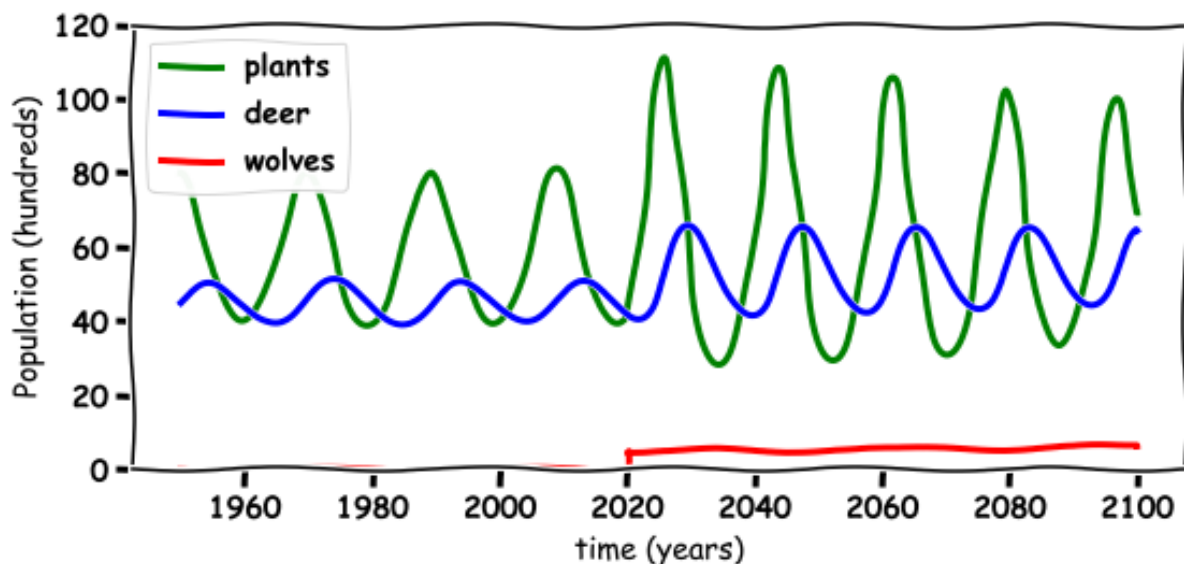
## Exercise 11

Label the graph arrows in the graph.



## Exercise 12:

Describe what is happening in the below graph:



### ANSWER:

When the Wolves were reintroduced, they scared the deer from the river which helped the plants grow. This also made more food for the deer.



## Discussion Questions

1. What happened to the environment when the wolves are re-introduced?
2. What happened to the wolf population when they are re-introduced?
3. What happened to the deer population when the wolves are re-introduced?
4. Do you think the wolves should have been re-introduced in Ireland?
5. What else could you model with these Equations?